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END TERM EXAMINATION

SECOND SEMESTER [B.Tech] MAY-JUNE 2026

Paper Code: ES-102**Subject: Programming in C****Time: 3 Hours****Maximum Marks: 60**

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit.

Q. No.	Question	Marks
Q1	Answer the following Questions (any five):	(5×4=20)
(a)	Explain the basic structure of a C program with an example?	(4)
(b)	Write a short note on data types in C.?	(4)
(c)	What are one-dimensional and two-dimensional arrays? Give examples.	(4)
(d)	Explain the use of any two string handling functions in C.	(4)
(e)	What is dynamic memory allocation? Mention any two related functions.	(4)
(f)	What are command line arguments? Give an example.	(4)
(g)	Explain the use of string.h with an example.	(4)
UNIT - I		
Q2 (a)	Write a C program that accepts a number from the user and checks whether it is even or odd. Explain the logic and use of conditional operator in your program.	(4)
(b)	Discuss different types of operators in C with suitable examples (arithmetic, relational, logical, and assignment).	(3)
(c)	Write a program in C to accept two integers and display their sum and product.	(3)
Q3 (a)	Explain the purpose of using flowcharts and pseudocode.	(4)
(b)	What are increment and decrement operators? Give examples.	(3)
(c)	What is the role of the preprocessor in a C program?	(3)
UNIT - II		
Q4 (a)	Describe storage classes in C with examples.	(3)
(b)	Write a recursive function to calculate the factorial of a given number.	(3)
(c)	Write a C program that accepts a 2D array of integers (3×3), calculates the sum of all elements, and displays the array and sum. Explain the use of nested loops in your program.	(4)
Q5 (a)	Write a C program to find the sum of even numbers from 1 to 100 using a loop.	(3)
(b)	Write a C program to accept a string from the user and count the number of vowels.	(3)
(c)	Write a note on the following: i. Call by value ii. Call by reference	(4)
UNIT - III		
Q6 (a)	Create a structure Book with fields: title, author, and price. Write a C program to read data of 5 books, store them in an array of structures, and display the books with price > ₹500. Include functions for input and display.	(4)
(b)	Write a program to open a file in write mode, take input from the user, and save it into the file.	(3)
(c)	What are pointers to functions? Write a program to demonstrate it.	(3)
Q7 (a)	Define a structure named Student with fields roll number, name, and marks. Write a C program to read and display information of 3 students using structures.	(4)

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Q. No.	Question	Marks
(b)	Write a C program that uses a pointer to display the address and value of an integer variable.	(3)
(c)	Explain the scope and life of a variable with an example.	(3)
UNIT - IV		
Q8 (a)	Write a C program to find the Fibonacci series up to n terms.	(4)
(b)	Implement bubble sort in C with suitable comments.	(3)
(c)	Explain the purpose of ctype.h with any two functions.	(3)
Q9 (a)	Write a C program to calculate the square root of a number using math.h ?	(4)
(b)	Describe the working of Insertion Sort with an example array.	(3)
(c)	Write a complete C program to accept an array of 10 integers, reverse the array, and display both the original and reversed arrays. Mention which header files you included and why.	(3)

**** End of Question Paper ****